Identification and CCP Estimation

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- Suppose that each period $t \in \{1, \dots, T\}$, the agent observes the realization (x_t, ϵ_t), and chooses d_t to sequentially maximize:

$$E\left\{\sum_{\tau=t}^{T} \sum_{j=1}^{J} \beta^{\tau-1} d_{j\tau}\left[u_{j\tau}(x_\tau)+\epsilon_{j\tau}\right] \middle| x_t, \epsilon_t\right\} \tag{1}$$

where $\beta \in (0,1)$ is a discount factor and:
  - $t \in \{1, \dots, T\}$ stands for time and $T \leq \infty$ denotes the time horizon.
  - $x_t \in \{1, \dots, X\} \equiv \mathbb{X}$ is the state at t, where $\mathbb{X}$ is a finite set.
  - $j \in \{1, \dots, J\}$ is a (mutually exclusive) choice.
  - $d_{jt} = 1$ if j is picked at t, otherwise $d_{jt} = 0$ and $d_t \equiv (d_{1t}, \dots, d_{Jt})$.
  - $\epsilon_t \equiv (\epsilon_{1t}, \dots, \epsilon_{Jt})$ where $\epsilon_{jt} \in \mathbb{R}$ with continuous support for all (j,t) and $E\left[\max\left\{\epsilon_{1t}, \dots, \epsilon_{Jt}\right\} \middle| x_t\right] \leq \overline{\epsilon} < \infty$.
  - $f_{jt}(x_{t+1}|x_t)$ is the probability of x_{t+1} occurring in period $t+1$ conditional on x_t and $d_{jt} = 1$.
  - conditional independence holds, meaning:

$$g_{t,j,x,\epsilon}\left(x_{t+1}, \epsilon_{t+1} \middle| x_t, \epsilon_t\right) = g_{t+1}\left(\epsilon_{t+1}|x_{t+1}\right) f_{jt}(x_{t+1}|x_t)$$

where $g_t\left(\epsilon_t|x_t\right)$ has continuous support.

- Denote the optimal decision rule at t as $d_t^o(x_t, \epsilon_t)$, with j^{th} element $d_{jt}^o(x_t, \epsilon_t)$ and define the *ex-ante* or *social surplus function* as:

$$V_t(x_t) = E\left\{\sum_{\tau=t}^{T}\sum_{j=1}^{J}\beta^{\tau-t-1}d_{j\tau}^o\left(x_\tau, \epsilon_\tau\right)\left(u_{j\tau}(x_\tau) + \epsilon_{j\tau}\right)|x_t\right\}$$

- The *choice specific* or *conditional value* function for action j is defined by:

$$v_{jt}(x_t) = u_{jt}(x_t) + \beta\sum_{x_{t+1}=1}^{X}V_{t+1}(x_{t+1})f_{jt}\left(x_{t+1}|x_t\right) \tag{2}$$

- Integrating $d_{jt}^o(x_t, \epsilon)$ over $\epsilon \equiv (\epsilon_1, \ldots, \epsilon_J)$ the CCPs are defined by:

$$p_{jt}(x_t) \equiv E\left[d_{jt}^o\left(x_t, \epsilon\right)|x_t\right] \equiv \int d_{jt}^o\left(x_t, \epsilon_t\right)g_t\left(\epsilon_t\right)d\epsilon_t \tag{3}$$

- Let $p_t(x_t) \equiv (p_{1t}(x_t), \ldots, p_{Jt}(x_t))$.

Inversion

Each CCP is a mapping of differences in the conditional valuation functions

- For convenience, drop the t subscript from x_t and ϵ_{jt} and define:

$$\begin{aligned} \Delta v_{jt}(x) &\equiv v_{jt}(x) - v_{Jt}(x) \\ G_{jt}(\epsilon | x) &\equiv \partial G_t(\epsilon | x) / \partial \epsilon_j \end{aligned}$$

where $G_t(\epsilon | x)$ is the CDF of $g_t(\epsilon | x)$. Then:

$$\begin{aligned} p_{jt}(x) &\equiv \int d_{jt}^o(x, \epsilon) g_t(\epsilon | x) d\epsilon \\ &= \int I\{\epsilon_k \leq \epsilon_j + \Delta v_{jt}(x) - \Delta v_{kt}(x) \forall\ k \neq j\} g_t(\epsilon | x) d\epsilon \\ &= \int_{\epsilon_j=-\infty}^{\infty} \int_{-\infty}^{\epsilon_j + \Delta v_{jt}(x) - \Delta v_{1t}(x)} \ldots \int_{-\infty}^{\epsilon_j + \Delta v_{jt}(x) - \Delta v_{J-1,t}(x)} \int_{-\infty}^{\epsilon_j + \Delta v_{jt}(x)} g_t(\epsilon | x) d\epsilon \\ &= \int_{-\infty}^{\infty} G_{jt}\left(\begin{array}{c} \epsilon_j + \Delta v_{jt}(x) - \Delta v_{1t}(x), \ldots \\ \ldots, \epsilon_j, \ldots, \epsilon_j + \Delta v_{jt}(x) \end{array} \bigg| x \right) d\epsilon_j \end{aligned}$$

Inversion

There are as many CCPs as there are conditional valuation functions

- For any vector $J-1$ dimensional vector $\delta \equiv (\delta_1, \ldots, \delta_{J-1})$ define:

$$Q_{jt}\left(\delta, x\right) \equiv \int_{-\infty}^{\infty} G_{jt}\left(\epsilon_j + \delta_j - \delta_1, \ldots, \epsilon_j, \ldots, \epsilon_j + \delta_j \left| x \right.\right) d\epsilon_j$$

- We interpret $Q_{jt}\left(\delta, x\right)$ as the probability taking action j in a static random utility model (RUM) where the payoffs are $\delta_j + \epsilon_j$ and the probability distribution of disturbances is given by $G_t\left(\epsilon \left| x \right.\right)$.
- It follows from the definition of $Q_{jt}\left(\delta, x\right)$ that:

$$0 \leq Q_{jt}\left(\delta, x\right) \leq 1 \text{ for all } \left(j, t, \delta, x\right) \text{ and } \sum_{j=1}^{J-1} Q_{jt}\left(\delta, x\right) \leq 1$$

- In particular the previous slide implies that for any given (j, t, x):

$$p_{jt}(x) = \int_{-\infty}^{\infty} G_{jt}\left(\begin{array}{c} \epsilon_j + \Delta v_{jt}(x) - \Delta v_{1t}(x), \\ \ldots, \epsilon_j, \ldots, \epsilon_j + \Delta v_{jt}(x) \end{array} \left| x \right.\right) d\epsilon_j \equiv Q_{jt}\left(\Delta v_t(x), x\right)$$

Inversion

Proposition 1 of Hotz and Miller (1993)

Theorem (Inversion)

For each (t, δ, x) *define:*

$$Q_t(\delta, x) \equiv \left(Q_{1t}(\delta, x), \ldots Q_{J-1,t}(\delta, x)\right)'$$

Then the vector function $Q_t(\delta, x)$ *is invertible in* δ *for each* (t, x).

- Note that $p_{Jt}(x) = Q_{Jt}(\Delta v_t, x)$ is a linear combination of the other equations in the system because $\sum_{k=1}^{J} p_k = 1$.
- Let $p \equiv (p_1, \ldots, p_{J-1})$ where $0 \leq p_j \leq 1$ for all $j \in \{1, \ldots, J-1\}$ and $\sum_{j=1}^{J-1} p_j \leq 1$. Denote the inverse of $Q_{jt}(\Delta v_t, x)$ by $Q_{jt}^{-1}(p, x)$.
- The inversion theorem implies:

$$\begin{bmatrix} \Delta v_{1t}(x) \\ \vdots \\ \Delta v_{J-1,t}(x) \end{bmatrix} = \begin{bmatrix} Q_{1t}^{-1}\left[p_t(x), x\right] \\ \vdots \\ Q_{J-1,t}^{-1}\left[p_t(x), x\right] \end{bmatrix}$$

Inversion

Interpreting the inversion expression

- $Q_{jt}^{-1}(p, x)$ has an intuitive interpretation:
  - Given x and $p(x)$ the agent is indifferent between the j^{th} and J^{th} choices for values of ϵ'_j and ϵ'_J satisfying:

$$\begin{aligned} v_{jt}(x) + \epsilon'_j &= v_{Jt}(x) + \epsilon'_J \\ \Rightarrow \Delta v_{jt}(x) &= \epsilon'_J - \epsilon'_j \\ &= Q_{jt}^{-1}\left[p_t(x), x\right] \end{aligned}$$

  - Thus the value of $Q_{jt}^{-1}\left[p_t(x), x\right]$ is the difference between the j^{th} and J^{th} taste shocks that would make the agent indifferent between those two choices.
  - More generally the value of the vector mapping:

$$Q_t^{-1}\left[p_t(x), x\right] = \left(Q_{1t}^{-1}\left[p_t(x), x\right], \ldots, Q_{J-1t}^{-1}\left[p_t(x), x\right]\right)$$

    corresponds to the value of a vector $\epsilon' \equiv \left(\epsilon'_1, \ldots, \epsilon'_J\right)$ that renders the agent indifferent to all the choices.

Inversion

Using the inversion theorem

- The inversion theorem exploits conditional independence to finesse optimization and integration.
- More specifically we use the inversion theorem to:
  1. provide empirically tractable *representations of the conditional value functions*.
  2. analyze *identification* in dynamic discrete choice models.
  3. provide convenient parametric forms for the density of ϵ_t *generalizing T1EV*.
  4. generalize the renewal and terminal state properties often used in empirical work to *finite dependence*, by obtaining restrictions on the state variable transitions used to implement CCP estimators.
  5. introduce methods for *incorporating unobserved state variables* into CCP estimation.

- Define the conditional value function correction as:

$$\psi_{jt}(x) \equiv V_t(x) - v_{jt}(x)$$

- Suppose that instead of taking the optimal action she committed to taking action j instead. Then the expected lifetime utility would be:

$$v_{jt}(x) + E_t[\epsilon_j | x]$$

so committing to j before ϵ_t is revealed entails a loss of:

$$V_t(x) - v_{jt}(x) - E_t[\epsilon_j | x] = \psi_{jt}(x) - E_t[\epsilon_j | x]$$

- For example if $E_t[\epsilon | x] = 0$, the loss simplifies to $\psi_{jt}(x)$.

- From their respective definitions:

$$
\begin{aligned}
& V_t(x) - v_{it}(x) \\
= \; & \sum_{j=1}^{J} \left\{ p_{jt}(x) \left[v_{jt}(x) - v_{it}(x) \right] + \int \epsilon_{jt} d_{jt}^{o}\left(x, \epsilon\right) g_t\left(\epsilon \left| x \right.\right) d\epsilon \right\}
\end{aligned}
$$

- But:

$$
v_{jt}(x) - v_{it}(x) = Q_{jt}^{-1}\left[p_t(x), x\right] - Q_{it}^{-1}\left[p_t(x), x\right]
$$

and

$$
\begin{aligned}
& \int \epsilon_j d_{jt}^{o}\left(x, \epsilon\right) g\left(\epsilon \left| x \right.\right) d\epsilon \\
= \; & \int \prod_{k=1}^{J} 1 \left\{ \begin{array}{l} \epsilon_k - \epsilon_j \\ \leq Q_{jt}^{-1}\left[p_t(x), x\right] - Q_{kt}^{-1}\left[p_t(x), x\right] \end{array} \right\} \epsilon_{jt} g_t\left(\epsilon \left| x \right.\right) d\epsilon
\end{aligned}
$$

- Therefore $\psi_{it}\left(x\right) \equiv V_t(x) - v_{it}(x)$ is identified if $G_t\left(\epsilon \left| x \right.\right)$ is known and (x, d) is the DGP.

Conditional Value Function Correction

Correction factor for extended nested logit

Lemma

For the nested logit $G(\epsilon)$:

$$\psi_j(p) = \gamma - \sigma \ln(p_j) - (1-\sigma)\ln\left(\sum_{k \in \mathcal{J}} p_k\right)$$

- Note that $\psi_j(p)$ only depends on the conditional choice probabilities for choices that are in the nest: the expression is the same no matter how many choices are outside the nest or how those choices are correlated.
- Hence, $\psi_j(p)$ will only depend on $p_{j'}$ if ϵ_j and $\epsilon_{j'}$ are correlated. When $\sigma = 1$, ϵ_j is independent of all other errors and $\psi_j(p)$ only depends on p_j.

Conditional Valuation Function Representation

Telescoping one period forward

- From its definition:

$$v_{jt}(x) = u_{jt}(x) + \beta \sum_{x'=1}^{X} V_{t+1}(x') f_{jt}\left(x'|x\right)$$

- Substituting for $V_{t+1}(x')$ using conditional value function correction we obtain:

$$v_{jt}(x) = u_{jt}(x) + \beta \sum_{x'=1}^{X} \left[v_{1,t+1}(x') + \psi_{1,t+1}(x')\right] f_{jt}\left(x'|x\right)$$

- We could repeat this procedure ad infinitum, substituting in for $v_{1,t+1}(x')$ by using the definition for $\psi_{1t}(x)$.

Conditional Valuation Function Representation

Special case of Theorem 1 of Arcidiacono and Miller (2011)

Theorem (Representation)

Define $\kappa_t(x'|x,j) \equiv f_{jt}(x'|x)$ *and for* $\tau = t+1, \ldots, T-1$*:*

$$\kappa_\tau(x_{\tau+1}|x,j) \equiv \sum_{x_\tau=1}^{X} f_{1\tau}(x_{\tau+1}|x_\tau)\kappa_{\tau-1}(x_\tau|x,j)$$

For any (t,x,j)*:*

$$\begin{aligned} v_{jt}(x) &= u_{jt}(x) \\ &+ \sum_{\tau=t+1}^{T}\sum_{x_\tau=1}^{X} \beta^{\tau-t}\left\{u_{1\tau}(x_\tau) + \psi_1[p_\tau(x_\tau)]\right\}\kappa_{\tau-1}(x_\tau|x,j) \end{aligned}$$

- This expression for $v_{jt}(x)$ dispenses with recursive maximization.
- The solution to the optimization problem is embedded in the CCPs.

- The optimization model is fully characterized by the time horizon, the utility flows, the discount factor, the transition matrix of the observed state variables, and the distribution of the unobserved variables, summarized with the notation (T, β, f, g, u).
- The data comprise observations for a real or synthetic panel on the observed part of the state variable, x_t, and decision outcomes, d_t.
- Following most of the empirical work in this area we consider identification when (T, β, f, g) are assumed to be known.
- Thus the goal is to identify u from (x_t, d_t) when (T, β, f, g) is known.

Identifying the Primitives

Observational Equivalence

- It is widely believed that u is only identified relative to one choice per period for each state.
- Can we say more than that?
- For each (x, t) let $l(x, t) \in \{1, \ldots, J\}$ denote any arbitrarily defined normalizing action.
- Let $c_t(x) \in \mathbb{R}$ its associated benchmark flow utility, meaning $u^*_{l(x,t),t}(x) \equiv c_t(x)$.
- Assume $\{c_t(x)\}_{t=1}^{T}$ is bounded for each $x \in \{1, \ldots, X\}$.

Theorem (Observational Equivalence, Arcidiacono and Miller, 2020)

For each $R \in \{1, 2, \ldots\}$, define for all $x \in \{1, \ldots, X\}$, $j \in \{1, \ldots, J\}$ and $t \in \{1, \ldots, R\}$:

$$u_{jR}^{*}(x) \equiv u_{jR}(x) + c_{R}(x) - u_{l(x,R),R}(x) \quad (4)$$

$$u_{jt}^{*}(x) \equiv u_{jt}(x) + c_{t}(x) - u_{l(x,t),t}(x) + \sum_{\tau=t+1}^{R} \sum_{x'=1}^{X} \beta^{\tau-t} \left\{ \begin{array}{l} \left[c_{\tau}(x') - u_{l(x,\tau),\tau}(x') \right] \times \\ \left[\kappa_{\tau-1}(x'|x_{t}, l(x,t)) - \kappa_{\tau-1}(x'|x_{t}, j) \right] \end{array} \right\} \quad (5)$$

(T, β, f, g, u^{*}), is observationally equivalent to (T, β, f, g, u) in the limit of $R \rightarrow T$. Conversely suppose (T, β, f, g, u^{*}) is observationally equivalent to (T, β, f, g, u). For each date and state select any action $l(x,t) \in \{1, \ldots, J\}$ with payoff $u_{l(x,t),t}^{*}(x) \equiv c_{t}(x) \in \mathbb{R}$. Then (4) and (5) hold for all (t, x, j).

Theorem (Identification)

For all j, t, and x:

$$\begin{aligned} u_{jt}(x) &= u_{1t}(x) + \psi_{1t}(x) - \psi_{jt}(x) \\ &+ \sum_{\tau=t+1}^{T} \sum_{x_\tau=1}^{X} \beta^{\tau-t} \left\{ \begin{array}{l} [u_{1\tau}(x_\tau) + \psi_{1t}(x_\tau)] \times \\ [\kappa_{\tau-1}(x_\tau|x,1) - \kappa_{\tau-1}(x_\tau|x,j)] \end{array} \right\} \end{aligned} \tag{6}$$

- If (T, β, f, g) is known, and if a payoff, say the first, is also known for every state and time, then u is identified.

- By the Law of Large Numbers the cell estimators $\widehat{f}_{jt}\left(x' \left|x\right.\right)$ and $\widehat{p}_{jt}\left(x\right)$ converge to their population analogues
- By the Central Limit Theorem both estimators converge at $\sqrt{N}$ and and have asymptotic normal distributions.
- Both $\widehat{f}_{jt}\left(x' \left|x\right.\right)$ and $\widehat{p}_{jt}\left(x\right)$ are ML estimators for $f_{jt}\left(x' \left|x\right.\right)$ and $p_{jt}(x)$ and obtain the Cramer-Rao lower bound asymptotically.
- Since and $u_{jt}(x)$ is exactly identified, it follows by the *invariance principle* that $\widehat{u}_{jt}(x)$ is consistent and asymptotically efficient for $u_{jt}(x_t)$, also attaining its Cramer Rao lower bound.
- Greater efficiency can only be obtained by making functional form assumptions about $u_{jt}(x_t)$ and $f_{jt}\left(x' \left|x\right.\right)$.

Restricting the Parameter Space

Parameterizing the primitives

- In practice applications further restrict the parameter space.
- For example assume $\theta \equiv \left(\theta^{(1)}, \theta^{(2)}\right) \in \Theta$ is a closed convex subspace of Euclidean space, and:
  - $u_{jt}(x) \equiv u_j(x, \theta^{(1)})$
  - $f_{jt}(x|x_{nt}) \equiv f_{jt}(x|x_{nt}, \theta^{(2)})$
- We now define the model by (T, β, θ, g).
- Assume the DGP comes from (T, β, θ_0, g) where:

$$\theta_0 \equiv \left(\theta_0^{(1)}, \theta_0^{(2)}\right) \in \Theta^{(interior)}$$

- For example many applications assume:
  - $u_{jt}(x) \equiv x'\theta_j^{(1)}$ is linear in x and does not depend on t
  - $f_{jt}(x|x_{nt})$ is degenerate, x following a deterministic law of motion that does not depend on t.

- A Quasi Maximum Likelihood (QML) estimator can be obtained by estimating:
  1. $\theta_0^{(2)}$ with $\theta_{LIML}^{(2)}$ from the data on $f_{jt}\left(x|x_t,\theta^{(2)}\right)$.
  2. $\kappa_\tau(x|t,x_t,k,\theta_0^{(2)})$ with $\widehat{\kappa}_\tau(x|t,x_t,k,\theta_{LIML}^{(2)})$ using $f_{jt}\left(x|x_t,\theta_{LIML}^{(2)}\right)$.
  3. $\psi_{1t}(x)$ with $\widehat{\psi}_{1t}(x)$ by substituting cell estimators $\widehat{p}_{jt}(x)$ for $p_{jt}(x)$.
  4. $v_{jt}\left(x,\theta^{(1)},\theta_0^{(2)}\right)$ with $\widehat{v}_{jt}\left(x,\theta^{(1)},\theta_{LIML}^{(2)}\right)$ for any given $\theta^{(1)}$, given below.
  5. $p_{jt}\left(x,\theta^{(1)},\theta_0^{(2)}\right)$ with $\widehat{p}_{jt}\left(x,\theta^{(1)},\theta_{LIML}^{(2)}\right)$ by substituting $\widehat{v}_{jt}\left(x,\theta^{(1)},\theta_{LIML}^{(2)}\right)$ for $v_{jt}\left(x,\theta^{(1)},\theta_0^{(2)}\right)$ in ML estimator.

- With respect to the last two steps:
  4. Appealing to the Representation theorem:

$$\widehat{v}_{jt}\left(x, \theta^{(1)}, \theta^{(2)}_{LIML}\right) = u_{jt}(x, \theta^{(1)}) + \widehat{h}_{jt}(x)$$

  where the numeric *dynamic correction factor* $\widehat{h}_{kt}(x)$ is defined:

$$\widehat{h}_{jt}(x) \equiv \sum_{\tau=t+1}^{T} \sum_{x_\tau=1}^{X} \beta^{\tau-t} \widehat{\psi}_{1\tau}(x_\tau) \widehat{\kappa}_{\tau-1}(x_\tau | t, x, j, \theta^{(2)}_{LIML})$$

  5. In T1EV applications:

$$\widehat{p}_{jt}(x, \theta^{(1)}, \theta^{(2)}_{LIML}) = \frac{\exp\left[u_{jt}(x, \theta^{(1)}) + \widehat{h}_{jt}(x)\right]}{\sum_{k=1}^{J} \exp\left[u_{kt}(x, \theta^{(1)}) + \widehat{h}_{kt}(x)\right]}$$

Minimum Distance Estimators (Altug and Miller,1998)

Minimizing the difference between unrestricted and restricted current payoffs

- Another approach is to match up the parametrization of $u_{jt}(x_t)$, denoted by $u_{jt}(x_t, \theta^{(1)})$, to its representation as closely as possible:
  1. Form the vector function where $\Psi(p, f)$ by stacking:

$$\begin{aligned}\Psi_{jt}(x_t, p, f) \equiv\ & \psi_{1t}(x_t) - \psi_{jt}(x_t) \\ & + \sum_{\tau=1}^{T-t} \sum_{x=1}^{X} \beta^{\tau} \psi_{1,t+\tau}(x) \left[\begin{array}{c} \kappa_{kt,\tau-1}(x|x_t) \\ -\kappa_{jt,\tau-1}(x|x_t) \end{array} \right]\end{aligned}$$

  2. Estimate the reduced form $\widehat{p}$ and $\widehat{f}$.
  3. Minimize the quadratic form to obtain:

$$\theta_{MD}^{(1)} = \underset{\theta^{(1)} \in \Theta^{(1)}}{\arg\min} \left[u(x, \theta^{(1)}) - \Psi\left(\widehat{p}, \widehat{f}\right) \right]' \widetilde{W} \left[u(x, \theta^{(1)}) - \Psi\left(\widehat{p}, \widehat{f}\right) \right]$$

  where $\widetilde{W}$ is a square $(J-1)\,TX$ weighting matrix.
- Note $\theta_{MD}^{(1)}$ has a closed form if $u(x; \theta_0^{(1)})$ is linear in $\theta_0^{(1)}$.

Simulated Moments Estimators

A simulated moments estimator (Hotz, Miller, Sanders and Smith, 1994)

- We could form a Methods of Simulated Moments (MSM) estimator from:
  1. Simulate a lifetime path from x_{nt_n} onwards for each j, using $\widehat{f}$ and $\widehat{p}$.
  2. Obtain estimates of $\widehat{E}\left[\epsilon_{jt}\middle|d_{jt}^{o}=1,x_t\right]$ from $\widehat{p}$.
  3. Stitch together a simulated lifetime utility outcome from the j^{th} choice at t_n onwards for n, to form $\widehat{v}_{jt}\left(x_{nt_n},\theta^{(1)},\widehat{f},\widehat{p}\right)$.
  4. Form the $J-1$ dimensional vector $m_n\left(x_{nt_n};\theta^{(1)},\widehat{f},\widehat{p}\right)$ from:

$$\begin{aligned} m_{nj}\left(x_{nt_n};\theta^{(1)},\widehat{f},\widehat{p}\right) \quad &\equiv \quad \widehat{v}_{jt_n}\left(x_{nt_n},\theta^{(1)},\widehat{f},\widehat{p}\right)-\widehat{v}_{Jt_n}\left(x_{nt_n},\theta^{(1)},\widehat{f},\widehat{p}\right) \\ &\quad +\widehat{\psi}_{jt}(x_{nt_n})-\widehat{\psi}_{Jt}(x_{nt_n}) \end{aligned}$$

  5. Given a weighting matrix W_S and an instrument vector z_n minimize:

$$N^{-1}\left[\sum_{n=1}^{N} z_n m_n\left(x_{nt_n};\theta^{(1)},\widehat{f},\widehat{p}\right)\right]' W_S\left[\sum_{n=1}^{N} z_n m_n\left(x_{nt_n};\theta^{(1)},\widehat{f},\widehat{p}\right)\right]$$

Finite Dependence
Motivation

- Estimation of dynamic discrete choice models is complicated by the calculation of expected future payoffs.
- These complications are mitigated when finite dependence holds.
- *Intuitively, ρ period dependence holds when two sequences of weighted (nonoptimal) choices leading off from different initial choices generate the same distribution of state variables $\rho + 1$ periods later.*
- Exploiting the finite dependence property reduces the number of CCPs to estimate.
- Most empirical applications have the finite dependence property.
- Under the conditional independence assumption, finite dependence has empirical content (can be tested) without specifying utilities.
- See Arcidiacono and Miller (2019, *Quantitative Economics*) for details.

- Define a weight sequence $\{\omega_{k\tau}(x_\tau, j, t, x)\}_{(k,\tau)=(1,t+1)}^{(J,t+\rho)}$ satisfying:

$$|\omega_{k\tau}(x_\tau, j, t, x)| < \infty \text{ and } \sum_{k=1}^{J} \omega_{k\tau}(x_\tau, j, t, x) = 1$$

- Analogous to the simpler case let $\kappa_{t+1}^{(\omega)}(x_{t+1}|j, t, x) \equiv f_{jt}(x_{t+1}|x)$ and:

$$\kappa_{\tau+1}^{(\omega)}(x_{\tau+1}|j, t, x) \equiv \sum_{x_\tau=1}^{X} \sum_{k=1}^{J} \omega_{k\tau}(x_\tau, j, t, x) f_{k\tau}(x_{\tau+1}|x_\tau) \kappa_\tau^{(\omega)}(x_\tau|j, t, x) \tag{7}$$

- Generalizing the telescoping argument given a few slides back:

$$v_{jt}(x) = V_t(x) - \psi_j[p_t(x)] = \tag{8}$$

$$\begin{cases} u_{jt}(x) + \sum_{x_{t+\rho+1}=1}^{X} \beta^{\rho+1} V_{t+\rho+1}(x_{t+\rho+1}) \kappa_{t+\rho+1}^{(\omega)}(x_{t+\rho+1} | j, t, x) \\ + \sum_{(k,\tau,x_\tau)=(1,t+1,1)}^{(J,t+\rho,X)} \beta^{\tau-t} \begin{bmatrix} u_{k\tau}(x_\tau) \\ +\psi_k[p_\tau(x_\tau)] \end{bmatrix} \begin{bmatrix} \omega_{k\tau}(x_\tau, j, t, x) \\ \times \kappa_\tau^{(\omega)}(x_\tau | j, t, x) \end{bmatrix} \end{cases}$$

- Interpreting this expression:
  - $V_{t+\rho+1}(x_{t+\rho+1})$ is the value upon reaching $x_{t+\rho+1}$.
  - $\kappa_{t+\rho+1}^{(\omega)}(x_{t+\rho+1} | j, t, x)$ is the probability of reaching $x_{t+\rho+1}$ by following the ω process from (t, x).
  - $\psi_k[p_\tau(x_\tau)]$ are correction terms for nonoptimal behavior.

Finite Dependence

Definition

- The pair of choices $\{i, j\}$ exhibits ρ*-period dependence* at (t, x) if there exist a pair of sequences of decision weights:

$$\{\omega_{k\tau}(x_\tau, i, t, x)\}_{(k,\tau)=(1,t+1)}^{(J,t+\rho)} \text{ and } \{\omega_{k\tau}(x_\tau, j, t, x)\}_{(k,\tau)=(1,t+1)}^{(J,t+\rho)}$$

  such that for all $x_{t+\rho+1} \in \{1, \ldots, X\}$:

$$\kappa_{t+\rho+1}^{(\omega)}(x_{t+\rho+1}|i, t, x) = \kappa_{t+\rho+1}^{(\omega)}(x_{t+\rho+1}|j, t, x)$$

- Finite dependence:
  1. trivially holds for $\rho = T - t$ when $T < \infty$, but only merits attention when $\rho < T - t$.
  2. extends to games by conditioning on the player as well.
  3. might hold for some choice pairs but not others, and for certain states but not others.
  4. could be defined for mixed choices to start the sequence, not just deterministic moves; this analysis extends to the more general case.

Finite Dependence

Representing utility

- If there is finite dependence for (i, j, t, x), then:

$$u_{jt}(x) + \psi_j[p_t(x)] - u_{it}(x) - \psi_i[p_t(x)] =$$

$$\sum_{(k,\tau,x_\tau)=(1,t+1,1)}^{(J,t+\rho,X)} \beta^{\tau-t} \left\{ \begin{array}{l} u_{k\tau}(x_\tau) \\ +\psi_k[p_\tau(x_\tau)] \end{array} \right\} \left[\begin{array}{l} \omega_{k\tau}(x_\tau, i, t, x)\kappa_\tau^{(\omega)}(x_\tau|i, t, x) \\ -\omega_{k\tau}(x_\tau, j, t, x)\kappa_\tau^{(\omega)}(x_\tau|j, t, x) \end{array} \right] \tag{9}$$

- To derive this equation:
  - replace $v_{jt}(x)$ with $V_t(x) - \psi_j[p_t(x)]$ in (8)
  - form the corresponding equation for i
  - difference the two resulting equations
  - note the $V_t(x)$ and $V_{t+\rho+1}(x_{t+\rho+1})\kappa_{t+\rho+1}^{(\omega)}(x_{t+\rho+1}|j, t, x)$ terms in (8) cancel with the terms in the corresponding expression for the i^{th} initial choice.

- A *terminal choice* ends the optimization problem or the game for that particular player.
- They are quite common in empirical applications.
- Supposing the first choice is terminal, then $u_{1t}(x)$ can be interpreted as the final payoff at (t, x) if that choice is made then.
- Setting $\omega_{k,t+1}(x_{t+1}, j, t, x) = 0$ for all (x_{t+1}, j), Equation (9) reduces to:

$$
\begin{aligned}
& u_{1t}(x) + \psi_1[p_t(x)] - u_{jt}(x) - \psi_j[p_t(x)] \\
= \quad & \textstyle\sum_{x_{t+1}=1}^{X} \beta \left\{ u_{1,t+1}(x_{t+1}) + \psi_1\left[p_{t+1}(x_{t+1})\right]\right\} f_{jt}(x_{t+1}|x)
\end{aligned}
$$

Simple Examples of Finite Dependence

Renewal choices are also quite common in empirical applications

- Similarly a *renewal choice* yields a probability distribution of the state variable next period that does not depend on the current state.
- If the first choice is a renewal choice, then for all $j \in \{1, \ldots, J\}$:

$$\begin{aligned} \sum_{x_{t+1}=1}^{X} f_{1,t+1}(x_{t+2}|x_{t+1}) f_{jt}(x_{t+1}|x) &= \sum_{x_{t+1}=1}^{X} f_{1,t+1}(x_{t+2}) f_{jt}(x_{t+1}|x) \\ &= f_{1,t+1}(x_{t+2}) \sum_{x_{t+1}=1}^{X} f_{jt}(x_{t+1}|x) \\ &= f_{1,t+1}(x_{t+2}) \end{aligned} \tag{10}$$

- In this case Equation (9) implies:

$$u_{1t}(x) + \psi_1[p_t(x)] - u_{jt}(x) - \psi_j[p_t(x)]$$

$$= \sum_{x_{t+1}=1}^{X} \beta \left\{ u_{1,t+1}(x_{t+1}) + \psi_1 \left[p_{t+1}(x_{t+1}) \right] \right\} \left[f_{jt}(x_{t+1}|x) - f_{1t}(x_{t+1}|x \right.$$

Simple Examples of Finite Dependence

Nonstationary search model

- Consider a simple search model in which all jobs are temporary, lasting only one period.
- Each period $t \in \{1, \ldots, T\}$ an individual may:
  - stay home by setting $d_{1t} = 1$
  - or seek work by applying for temporary employment setting $d_{2t} = 1$.
- Job applicants are successful with probability λ_t, time varying job offer arrival rates.
- Experience $x \in \{1, \ldots, X\}$ increases by one unit with each period of work, up to X, and does not depreciate.

- For all (t, x) with $x < X$ set:
  - $d_{1t} = 1$ (stay home) and then seek work with weight:

$$\begin{aligned} \lambda_t / \lambda_{t+1} &= \omega_{k=2,t+1}(x_{t+1} = x, j = 1, t, x) \\ &= 1 - \omega_{k=1,t+1}(x_{t+1} = x, j = 1, t, x) \end{aligned}$$

  - $d_{2t} = 1$ (seek work) and then stay home. So for $x_{t+1} \in \{x, x+1\}$:

$$\omega_{k=1,t+1}\left(x_{t+1}, j = 2, t, x\right) = 1$$

- This pair of sequencing attains one-period dependence because:

$$\kappa_2(x_{t+2} | j = 1, t, x) = \kappa_2(x_{t+2} | j = 2, t, x) = \begin{cases} 1 - \lambda_t \text{ for } x_{t+2} = x \\ \lambda_t \text{ for } x_{t+2} = x + 1 \end{cases}$$

- Note that if $\lambda_t > \lambda_{t+1}$ then $\omega_{k=2,t+1}(x, j = 1, t, x) > 1$ and:

$$\omega_{k=1,t+1}(x, j = 1, t) = 1 - \lambda_t / \lambda_{t+1} < 0$$

Relaxing the Conditional Independence Assumption

Motivating example

- Relaxing the conditional independence assumption is tantamount to reopening the multiple integration challenge.
- CCPs can be used in combination with the EM algorithm to estimate models of dynamic discrete choice when there is unobserved heterogeneity.
- Denote by:
  - $s_n \in \{1, 2, \ldots, S\}$ an *iid* random individual specific effect drawn from a finite distribution.
  - π_s the probability of being in unobserved state s.
  - $p_{(n,t)}$ future CCPs that are relevant for n at t.
  - $l(d_{nt}|x_{nt}, s_n, p, \theta)$ the quasi-likelihood of observing d_{nt} conditional on (x_{nt}, s_n) given θ and $p_{(n,t)}$.

Set initial values for $\theta^{(1)}$, $\pi^{(1)}$, and $p^{(1)}_{(n,t)}$ for each (n,t):

Step 1 Compute $q^{(m+1)}_{ns}$ as:

$$\begin{aligned} q^{(m+1)}_{ns} &= \Pr\left\{ s \,\middle|\, d_n, x_n, \theta^{(m)}, \pi^{(m)}_s, p^{(m)} \right\} \\ &= \frac{\pi^{(m)}_s \prod_{t=1}^{T} l\left[d_{nt} | x_{nt}, s, p^{(m)}_{(n,t)}, \theta^{(m)} \right]}{\sum_{s'=1}^{S} \pi^{(m)}_s \prod_{t=1}^{T} l\left(d_{nt} | x_{nt}, s', p^{(m)}_{(n,t)}, \theta^{(m)} \right)} \end{aligned}$$

Step 2 Compute $\pi^{(m+1)}_s$ according to:

$$\pi^{(m+1)}_s = \sum_{n=1}^{N} q^{(m+1)}_{ns} \,/\, N$$

Step 3 Update $p^{(m+1)}(x,s)$ using one of two rules below.

Step 4 Obtain $\theta^{(m+1)}$ as the solution to:

$$\arg\max_{\theta} \sum_{(n,s,t)=(1,1,1)}^{(N,S,T)} q^{(m+1)}_{ns} \ln\left[l\left(d_{nt} | x_{nt}, s_n, p^{(m+1)}_{(n,t)}, \theta \right) \right]$$

Relaxing the Conditional Independence Assumption

Updating the CCP's

- Take a weighted average of decisions to replace engine, conditional on x, where weights are the conditional probabilities of being in unobserved state s.

  Step 3A Update CCP's with:

$$p_j^{(m+1)}(x, s) = \frac{\sum_{n=1}^{N} \sum_{t=1}^{T} d_{jnt} q_{ns}^{(m+1)} I(x_{nt} = x)}{\sum_{n=1}^{N} \sum_{t=1}^{T} q_{ns}^{(m+1)} I(x_{nt} = x)}$$

- Use the quasi-likelihood:

  Step 3B Update CCP's with:

$$p^{(m+1)}(x_{nt}, s_n) = l(d_{nt1} = 1 | x_{nt}, s_n, p_{(n,t)}^{(m)}, \theta^{(m)})$$

Entry Exit Game
Choice variables

- Suppose there is a finite maximum number of firms in a market at any one time denoted by I.
- If a firm exits, the next period an opening occurs to a potential entrant, who may decide to exercise this one time option, or stay out.
- At the beginning of each period every incumbent firm has the option of quitting the market or staying one more period.
- Let $d_t^{(i)} \equiv \left(d_{t1}^{(i)}, d_{t2}^{(i)} \right)$, where $d_{t1}^{(i)} = 1$ means i exits or stays out of the market in period t, and $d_{t2}^{(i)} = 1$ means i enters or does not exit.
- If $d_{t2}^{(i)} = 1$ and $d_{t-1,1}^{(i)} = 1$ then the firm in spot i at time t is an entrant, and if $d_{t-1,2}^{(i)} = 1$ the spot i at time t is an incumbent.

Entry Exit Game
State variables

- In this application there are three components to the state variables and $x_t = (x_1, x_{2t}, s_t)$.
- The first is a permanent market characteristic, denoted by x_1, and is common across firms in the market. Each market faces an equal probability of drawing any of the possible values of x_1 where $x_1 \in \{1, 2, \ldots, 10\}$.
- The second, x_{2t}, is whether or not each firm is an incumbent, $x_{2t} \equiv \{d_{t-1,2}^{(1)}, \ldots, d_{t-1,2}^{(I)}\}$. Entrants pay a start up cost, making it more likely that stayers choose to fill a slot than an entrant.
- A demand shock $s_t \in \{1, \ldots, 5\}$ follows a first order Markov chain.
- In particular, the probability that $s_{t+1} = s_t$ is fixed at $\pi \in (0, 1)$, and probability of any other state occurring is equally likely:

$$\Pr\{s_{t+1} | s_t\} = \begin{cases} \pi \text{ if } s_{t+1} = s_t \\ (1-\pi)/4 \text{ if } s_{t+1} \neq s_t \end{cases}$$

- Each active firm produces one unit so revenue, denoted by y_t, is just price.
- Price is determined by:
  1. the supply of active firms in the market, $\sum_{i=1}^{I} d_{t2}^{(i)}$
  2. a permanent market characteristic, x_1
  3. the Markov demand shock s_t, which the researcher does not observe
  4. another temporary shock, denoted by η_t, distributed *iid* standard normal distribution, revealed to each market after the entry and exit decisions are made.
- The price (revenue) equation is:

$$y_t = \alpha_0 + \alpha_1 x_1 + \alpha_2 s_t + \alpha_3 \sum_{i=1}^{I} d_{t2}^{(i)} + \eta_t$$

Entry Exit Game

Expected profits conditional on competition

- We assume costs comprise a choice specific disturbance $\epsilon_{tj}^{(i)}$ that is privately observed, plus a linear function of $\left(x_t^{(i)}, s_t^{(i)}, d_t^{(-i)}\right)$.
- Net current profits for exiting incumbent firms, and potential entrants who do not enter, are $\epsilon_{1t}^{(i)}$. Thus $U_1^{(i)}\left(x_t^{(i)}, s_t^{(i)}, d_t^{(-i)}\right) \equiv 0$.
- Current profits from being active are the sum of $\left(\epsilon_{2t}^{(i)} + \eta_t\right)$ and:

$$U_2^{(i)}\left(x_t^{(i)}, s_t^{(i)}, d_t^{(-i)}\right) \equiv \theta_0 + \theta_1 x_1 + \theta_2 s_t + \theta_3 \sum_{\substack{i'=1 \\ i' \neq i}}^{I} d_{2t}^{(i')} + \theta_4 d_{1,t-1}^{(i)}$$

  where θ_4 is the startup cost that only entrants pay.
- In equilibrium $E\left(\eta_t\right) = 0$ so:

$$u_2^{(i)}\left(x_t, s_t\right) = \theta_0 + \theta_1 x_1 + \theta_2 s_t + \theta_3 \sum_{\substack{i'=1 \\ i' \neq i}}^{I} p_2^{(i')}\left(x_t, s_t\right) + \theta_4 d_{1,t-1}^{(i)}$$

- We assume the firm's private information, $\epsilon_{jt}^{(i)}$, is distributed T1EV.
- Exiting is a terminal choice, with a payoff normalized to zero.
- Given T1EV, the conditional value function for being active is:

$$
\begin{aligned}
v_2^{(i)}(x_t, s_t) \quad = \quad & u_2^{(i)}(x_t, s_t) \\
& -\beta \sum_{x \in X} \sum_{s \in S} \left(\ln \left[p_1^{(i)}(x, s) \right] \right) f_2^{(i)}(x, s | x_t, s_t)
\end{aligned}
$$

- We set the maximum number of firms in each market to 6 and the discount factor $\beta = 0.9$.
- Starting at an initial date with six potential entrants in the market, we solved the model, ran the simulations forward for 3,000 markets for 20 periods, using the last ten periods to estimate the model.

Entry Exit Game

Results from Monte Carlo simulations (Arcidiacono and Miller, 2011)

	DGP (1)	s_t Observed (2)	Ignore s_t (3)	CCP Model (4)	CCP Data (5)	Two-Stage (6)	No Prices (7)
Profit parameters							
θ_0 (intercept)	0	0.0207	−0.8627	0.0073	0.0126	−0.0251	−0.0086
		(0.0779)	(0.0511)	(0.0812)	(0.0997)	(0.1013)	(0.1083)
θ_1 (obs. state)	0.05	−0.0505	−0.0118	−0.0500	−0.0502	−0.0487	−0.0495
		(0.0028)	(0.0014)	(0.0029)	(0.0041)	(0.0039)	(0.0038)
θ_2 (unobs. state)	0.25	0.2529		0.2502	0.2503	0.2456	0.2477
		(0.0080)		(0.0123)	(0.0148)	(0.0148)	(0.0158)
θ_3 (no. of competitors)	−0.2	−0.2061	0.1081	−0.2019	−0.2029	−0.1926	−0.1971
		(0.0207)	(0.0115)	(0.0218)	(0.0278)	(0.0270)	(0.0294)
θ_4 (entry cost)	−1.5	−1.4992	−1.5715	−1.5014	−1.4992	−1.4995	−1.5007
		(0.0131)	(0.0133)	(0.0116)	(0.0133)	(0.0133)	(0.0139)
Price parameters							
α_0 (intercept)	7	6.9973	6.6571	6.9991	6.9952	6.9946	
		(0.0296)	(0.0281)	(0.0369)	(0.0333)	(0.0335)	
α_1 (obs. state)	−0.1	−0.0998	−0.0754	−0.0995	−0.0996	−0.0996	
		(0.0023)	(0.0025)	(0.0028)	(0.0028)	(0.0028)	
α_2 (unobs. state)	0.3	0.2996		0.2982	0.2993	0.2987	
		(0.0045)		(0.0119)	(0.0117)	(0.0116)	
α_3 (no. of competitors)	−0.4	−0.3995	−0.2211	−0.3994	−0.3989	−0.3984	
		(0.0061)	(0.0051)	(0.0087)	(0.0088)	(0.0089)	
π (persistence of unobs. state)	0.7			0.7002	0.7030	0.7032	0.7007
				(0.0122)	(0.0146)	(0.0146)	(0.0184)
Time (minutes)		0.1354	0.1078	21.54	27.30	15.37	16.92
		(0.0047)	(0.0010)	(1.5278)	(1.9160)	(0.8003)	(1.6467)

Mean and standard deviations for 100 simulations. Observed data consist of 3000 markets for 10 periods with 6 firms in each market. In column 7, the CCP's are updated with the model.